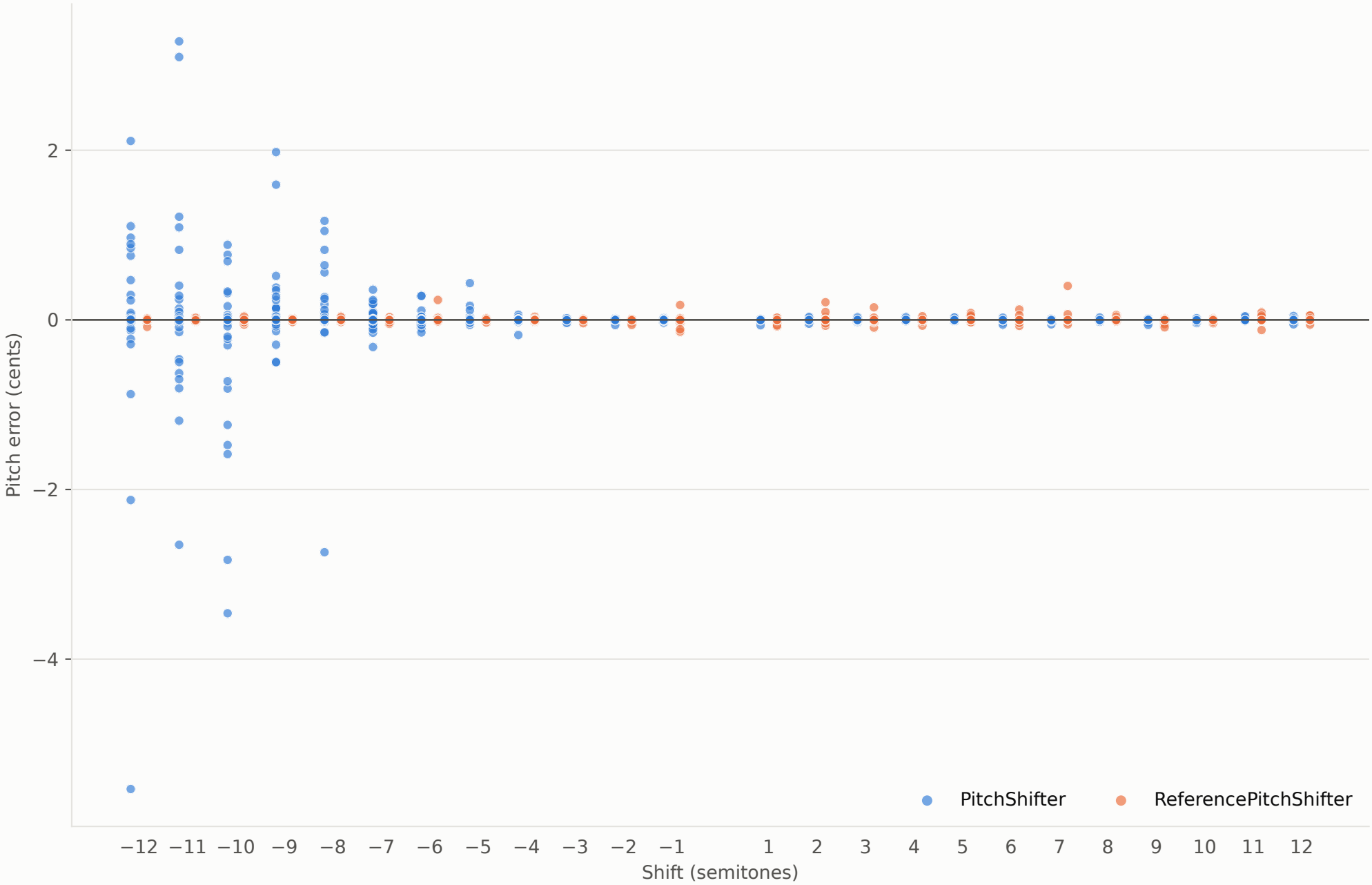
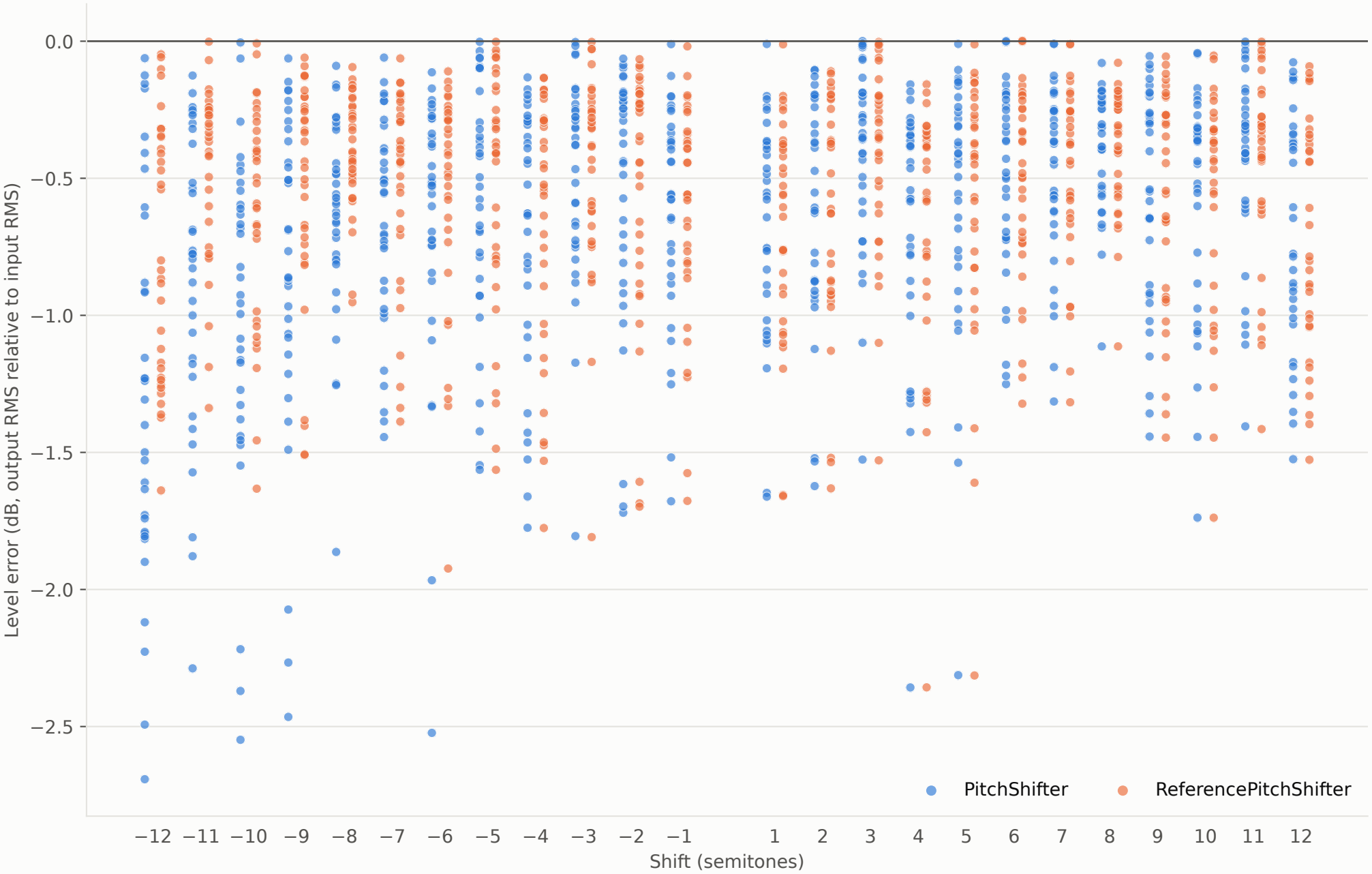
Test 1: Pitch shifter accuracy, pitch

Every tenor note (Ab2 to E5, written low Bb to high F#) through all 24 shifts, one dot per note. STK Saxofony input tuned to each target.



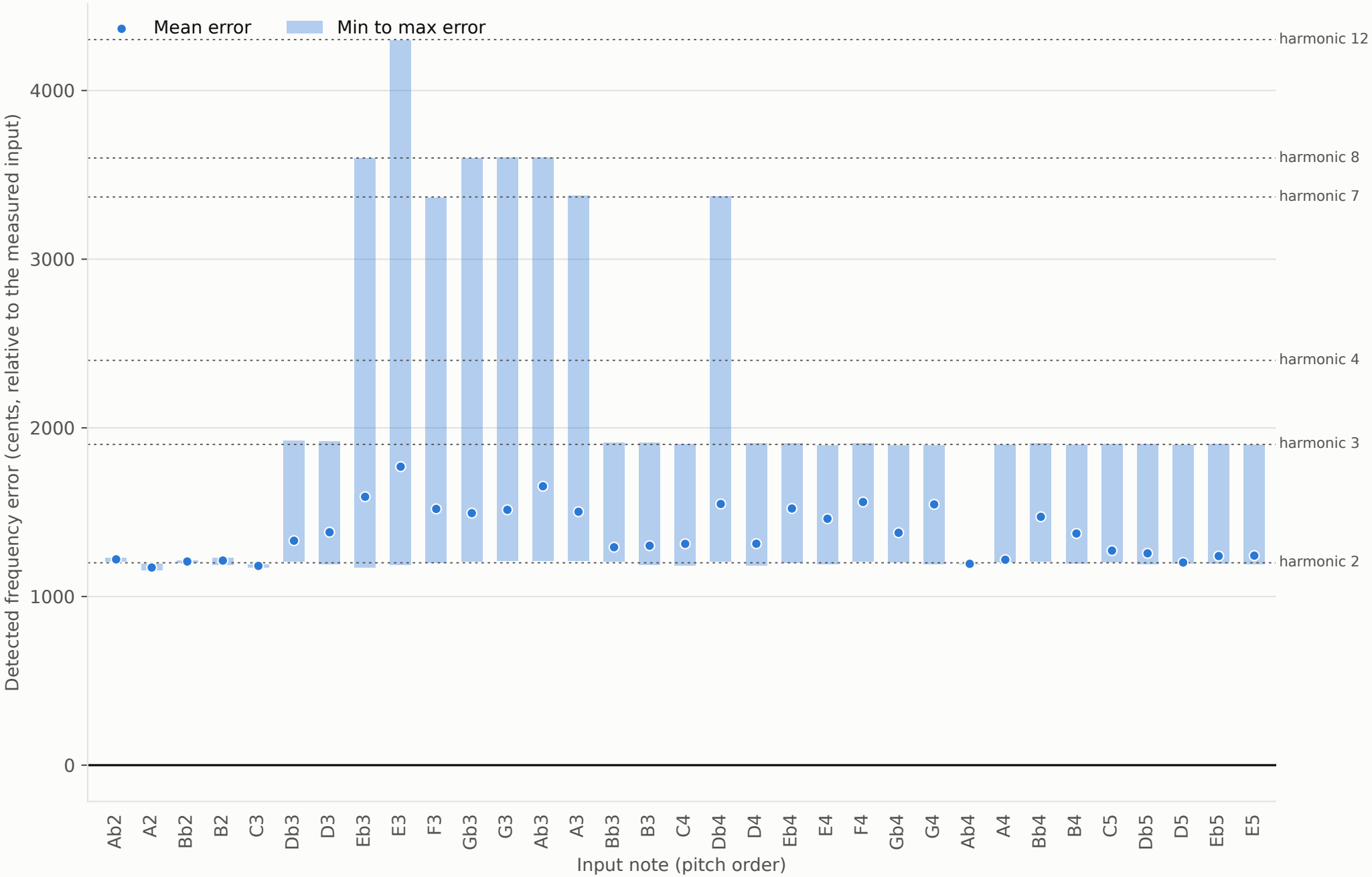
Test 1: Pitch shifter accuracy, level

Same runs as the previous page. RMS level of the settled output against the matching part of the input.



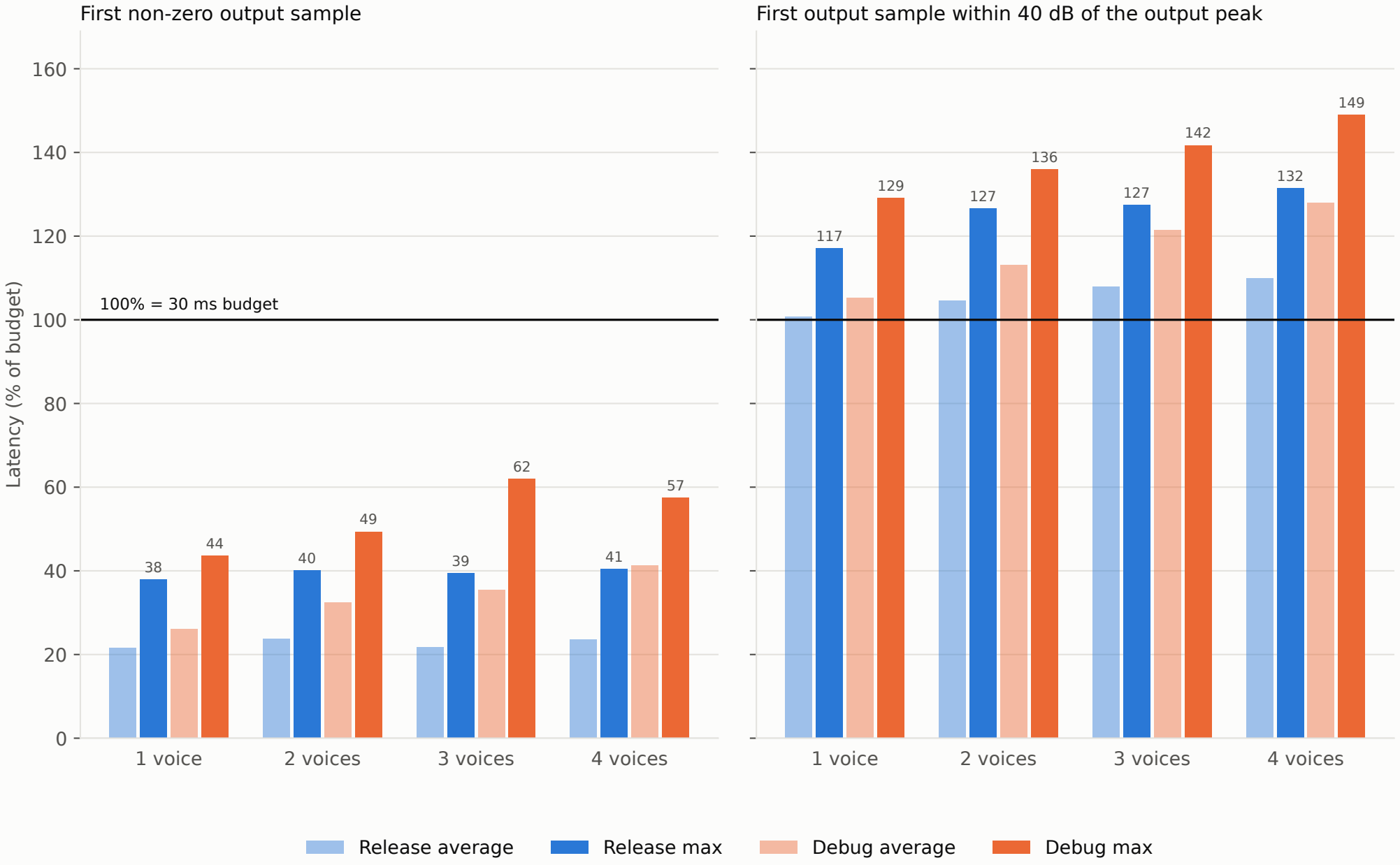
Test 2: Pitch detection accuracy

PitchDetector on STK Saxofony tones. 4921 estimates over 33 notes (8 trials per note); 4921 are more than 50 cents off.
Dashed lines mark where each harmonic falls. This tone's fundamental is 7 to 27 dB below its strongest harmonic (5 sampled notes)
and the detector reports the loudest peak. The output shifts do not depend on this detector (relative mode).



Test 3: End-to-end latency

50 random trials per voice count, identical in both builds: random note, shifts and block alignment; 100 ms STK snippet through HarmonizerEngine at 48 kHz, 512-sample blocks. Latency = sample delay + time to compute the delivering block.
Audio device and driver buffering is not included. The budget is a placeholder: change it with --budget-ms.
The first non-zero sample is a very quiet frame edge, well ahead of the audible sound; both are shown.



Test 4: Timbral quality, Ab2 (103.8 Hz)

Each harmonic's interpolated peak in the input against its expected position in the output (8192-sample Hann windows, averaged). Worst and mean are of absolute error across the harmonics of that shift; level is absolute, with no gain correction. Freq error covers found peaks only.

	PitchShifter					ReferencePitchShifter				
Shift	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found
-12	76.01	1.88	48.2	5.97	92/99	34.07	0.56	11.9	2.25	99/99
-11	128.41	2.52	54.2	5.16	94/99	21.00	0.41	11.4	1.93	96/99
-10	23.35	0.75	57.9	5.24	90/99	8.63	0.16	16.2	1.69	96/99
-9	88.36	1.42	54.5	4.68	95/99	15.69	0.32	13.3	1.56	98/99
-8	31.49	1.20	58.3	5.37	93/99	13.13	0.30	12.2	1.69	96/99
-7	15.29	0.72	46.7	4.20	96/99	28.56	0.44	11.9	1.54	97/99
-6	30.12	1.00	35.8	3.19	97/99	4.99	0.25	12.1	1.57	98/99
-5	86.33	1.36	37.7	2.85	97/99	9.69	0.27	10.2	1.71	98/99
-4	13.15	0.44	31.9	2.54	97/99	21.63	0.35	8.7	1.46	98/99
-3	8.40	0.21	21.9	1.60	98/99	31.31	0.69	10.4	1.54	99/99
-2	16.12	0.23	8.4	1.11	99/99	14.03	0.33	8.4	1.55	99/99
-1	3.03	0.08	3.6	0.88	99/99	11.41	0.34	8.2	1.71	99/99
+1	2.59	0.07	3.3	0.76	99/99	9.13	0.27	7.8	1.69	99/99
+2	1.11	0.07	3.4	0.75	99/99	18.00	0.44	9.7	1.82	99/99
+3	1.03	0.06	3.5	0.82	99/99	26.82	0.69	8.3	2.08	99/99
+4	0.76	0.07	2.8	0.80	99/99	33.96	0.81	8.4	2.03	99/99
+5	0.71	0.05	2.7	0.89	99/99	3.48	0.23	8.5	2.31	97/99
+6	0.72	0.06	5.4	0.97	99/99	6.28	0.23	11.9	2.39	98/99
+7	8.52	0.14	5.6	1.02	98/98	3.65	0.29	13.7	2.36	98/98
+8	1.71	0.07	3.7	1.11	98/98	20.33	0.55	13.1	2.66	98/98
+9	4.97	0.10	5.6	1.13	96/96	8.11	0.29	14.2	2.49	96/96
+10	2.16	0.06	3.9	1.12	94/94	3.89	0.24	14.4	2.57	94/94
+11	1.67	0.07	5.5	1.21	91/91	16.18	0.34	14.3	2.59	91/91
+12	0.64	0.06	7.5	1.32	87/87	2.83	0.21	16.0	2.53	87/87

Test 4: Timbral quality, E3 (164.8 Hz)

Each harmonic's interpolated peak in the input against its expected position in the output (8192-sample Hann windows, averaged). Worst and mean are of absolute error across the harmonics of that shift; level is absolute, with no gain correction. Freq error covers found peaks only.

	PitchShifter					ReferencePitchShifter				
Shift	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found
-12	222.03	4.33	35.6	5.17	82/84	1.04	0.06	6.7	1.17	84/84
-11	115.54	2.84	51.0	4.73	82/84	5.98	0.16	9.4	1.19	83/84
-10	17.03	0.83	35.7	4.20	82/84	2.92	0.14	8.2	1.38	84/84
-9	30.22	1.24	48.8	4.04	82/84	4.70	0.14	7.3	1.08	84/84
-8	34.19	1.31	47.1	3.78	84/84	3.39	0.19	7.7	1.33	84/84
-7	52.37	1.32	19.7	2.73	84/84	52.71	0.75	9.1	1.33	84/84
-6	69.94	1.43	20.3	2.21	84/84	3.83	0.11	8.4	1.22	84/84
-5	6.32	0.17	21.0	1.78	83/84	0.66	0.05	7.5	1.13	84/84
-4	0.79	0.09	6.3	1.18	84/84	4.74	0.12	8.5	1.33	84/84
-3	11.53	0.18	9.2	0.87	84/84	3.31	0.10	8.6	1.37	84/84
-2	0.90	0.04	4.4	0.69	84/84	2.24	0.08	8.7	1.44	84/84
-1	0.41	0.04	2.6	0.70	84/84	3.52	0.11	9.3	1.61	84/84
+1	0.40	0.04	3.8	0.70	84/84	10.00	0.22	8.6	1.60	84/84
+2	0.38	0.04	3.0	0.76	84/84	3.91	0.12	9.1	1.61	84/84
+3	0.74	0.04	2.9	0.75	84/84	3.11	0.15	9.2	1.79	84/84
+4	0.55	0.04	3.3	0.82	83/83	7.96	0.26	9.8	1.92	83/83
+5	0.67	0.04	2.8	0.82	81/81	1.67	0.08	9.6	1.91	81/81
+6	0.38	0.03	2.8	0.84	79/79	1.89	0.14	9.3	1.97	79/79
+7	0.36	0.04	3.0	0.89	76/76	2.93	0.15	9.0	1.98	76/76
+8	0.19	0.03	3.0	0.79	74/74	1.68	0.10	9.6	1.75	74/74
+9	0.28	0.03	2.7	0.85	71/71	1.77	0.11	10.0	1.90	71/71
+10	0.22	0.03	2.1	0.81	68/68	3.72	0.15	8.6	1.67	68/68
+11	0.36	0.03	3.0	0.90	64/64	5.34	0.24	9.0	2.13	64/64
+12	0.22	0.03	2.7	1.01	60/60	1.56	0.10	8.8	1.91	60/60

Test 4: Timbral quality, C4 (261.6 Hz)

Each harmonic's interpolated peak in the input against its expected position in the output (8192-sample Hann windows, averaged). Worst and mean are of absolute error across the harmonics of that shift; level is absolute, with no gain correction. Freq error covers found peaks only.

	PitchShifter					ReferencePitchShifter				
Shift	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found
-12	72.40	3.23	17.3	2.88	45/45	3.63	0.16	9.6	1.36	45/45
-11	41.81	2.24	17.4	2.26	45/45	4.15	0.21	8.1	1.26	45/45
-10	62.70	1.96	18.1	2.05	45/45	3.55	0.15	8.8	1.40	45/45
-9	9.54	0.60	10.0	1.75	45/45	3.44	0.19	8.9	1.29	45/45
-8	8.53	0.47	6.3	1.21	45/45	2.66	0.16	9.8	1.20	45/45
-7	5.74	0.29	5.7	1.19	45/45	13.75	0.56	9.4	1.40	45/45
-6	1.96	0.18	6.6	0.99	45/45	6.47	0.40	9.1	1.80	45/45
-5	0.55	0.07	3.4	0.86	45/45	3.37	0.17	8.9	1.38	45/45
-4	0.67	0.06	3.0	0.73	45/45	4.62	0.16	9.9	1.66	45/45
-3	0.28	0.04	1.9	0.56	45/45	2.78	0.10	7.7	1.07	45/45
-2	0.54	0.06	4.0	0.76	45/45	6.17	0.28	9.6	1.84	45/45
-1	0.33	0.04	2.7	0.65	45/45	0.67	0.05	8.7	1.63	45/45
+1	0.17	0.03	2.2	0.69	45/45	6.41	0.30	7.7	1.67	45/45
+2	0.36	0.03	2.8	0.76	45/45	5.74	0.34	8.8	1.90	45/45
+3	0.21	0.04	2.1	0.78	45/45	5.93	0.28	8.9	1.94	45/45
+4	0.24	0.04	2.8	0.87	45/45	1.38	0.09	8.6	2.08	45/45
+5	0.47	0.05	3.3	0.86	45/45	15.89	0.56	9.6	2.24	45/45
+6	0.37	0.03	2.3	0.86	44/44	3.17	0.12	8.5	1.79	44/44
+7	0.27	0.03	2.8	0.86	43/43	4.53	0.23	8.8	2.11	43/43
+8	0.29	0.04	3.1	0.85	42/42	1.71	0.09	9.8	2.08	42/42
+9	0.22	0.03	2.9	0.92	40/40	5.73	0.27	9.0	2.48	40/40
+10	0.19	0.03	2.3	0.90	39/39	1.54	0.11	9.2	2.25	39/39
+11	0.18	0.03	3.3	0.98	37/37	2.50	0.18	8.5	2.09	37/37
+12	0.14	0.03	2.4	1.10	37/37	3.33	0.17	8.1	2.41	37/37

Test 4: Timbral quality, Ab4 (415.3 Hz)

Each harmonic's interpolated peak in the input against its expected position in the output (8192-sample Hann windows, averaged). Worst and mean are of absolute error across the harmonics of that shift; level is absolute, with no gain correction. Freq error covers found peaks only.

	PitchShifter					ReferencePitchShifter				
Shift	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found
-12	0.86	0.16	3.4	0.90	29/29	1.01	0.10	5.6	1.13	29/29
-11	1.17	0.15	1.6	0.51	29/29	1.00	0.12	5.7	0.90	29/29
-10	1.23	0.16	1.7	0.59	29/29	1.77	0.15	5.4	0.92	29/29
-9	0.70	0.10	1.9	0.47	29/29	1.90	0.15	5.3	0.84	29/29
-8	0.58	0.09	1.2	0.40	29/29	0.90	0.11	4.9	0.83	29/29
-7	0.62	0.05	2.1	0.42	29/29	0.42	0.08	4.7	0.82	29/29
-6	0.61	0.07	1.6	0.43	29/29	0.79	0.11	2.8	0.83	29/29
-5	0.67	0.06	2.7	0.67	29/29	0.74	0.12	3.0	1.07	29/29
-4	0.33	0.05	2.1	0.53	29/29	2.90	0.19	7.1	1.05	29/29
-3	0.58	0.05	1.4	0.44	29/29	0.86	0.13	5.0	0.87	29/29
-2	0.15	0.03	2.6	0.48	29/29	0.57	0.08	3.2	0.85	29/29
-1	0.30	0.06	2.7	0.29	29/29	1.07	0.17	7.3	1.09	29/29
+1	0.26	0.04	2.8	0.67	29/29	1.22	0.15	3.3	1.00	29/29
+2	0.21	0.05	2.3	0.77	29/29	1.60	0.22	7.4	1.80	29/29
+3	0.50	0.06	2.1	0.73	29/29	1.23	0.18	6.3	1.41	29/29
+4	0.50	0.06	2.2	0.72	29/29	1.07	0.13	3.0	1.09	29/29
+5	0.84	0.08	2.2	0.87	29/29	1.97	0.23	7.3	1.82	29/29
+6	0.40	0.06	2.8	0.85	29/29	1.94	0.19	6.8	1.64	29/29
+7	0.53	0.07	2.2	0.96	29/29	0.78	0.19	6.4	2.07	29/29
+8	0.57	0.08	2.2	1.01	27/27	0.98	0.17	5.8	1.93	27/27
+9	0.59	0.07	2.9	1.29	27/27	1.26	0.32	7.6	3.13	27/27
+10	0.71	0.09	2.4	1.13	26/26	1.19	0.24	6.4	2.11	26/26
+11	0.74	0.08	2.7	1.27	24/24	1.12	0.20	7.4	1.92	24/24
+12	0.76	0.08	2.8	1.34	24/24	0.40	0.10	6.2	1.93	24/24

Test 4: Timbral quality, E5 (659.3 Hz)

Each harmonic's interpolated peak in the input against its expected position in the output (8192-sample Hann windows, averaged). Worst and mean are of absolute error across the harmonics of that shift; level is absolute, with no gain correction. Freq error covers found peaks only.

	PitchShifter					ReferencePitchShifter				
Shift	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found	Worst freq (cents)	Mean freq (cents)	Worst level (dB)	Mean level (dB)	Peaks found
-12	0.14	0.02	1.9	0.74	21/21	0.08	0.02	5.0	0.99	21/21
-11	0.62	0.05	1.6	0.50	21/21	5.00	0.25	9.1	1.23	21/21
-10	0.27	0.03	2.7	0.68	21/21	0.09	0.01	4.5	1.11	21/21
-9	0.64	0.04	1.4	0.44	21/21	1.27	0.07	7.9	0.80	21/21
-8	0.20	0.02	1.0	0.39	21/21	0.06	0.01	4.8	0.75	21/21
-7	0.28	0.04	1.8	0.60	21/21	6.41	0.31	8.2	0.99	21/21
-6	0.13	0.02	2.2	0.68	21/21	0.39	0.03	7.2	1.37	21/21
-5	0.12	0.03	2.0	0.65	21/21	0.13	0.02	4.3	1.13	21/21
-4	0.33	0.04	2.1	0.64	21/21	1.77	0.10	10.0	1.37	21/21
-3	0.48	0.05	1.4	0.54	21/21	4.90	0.37	8.7	1.51	21/21
-2	0.21	0.03	1.8	0.76	21/21	0.15	0.02	8.8	1.63	21/21
-1	0.16	0.03	1.9	0.56	21/21	4.73	0.24	7.9	1.17	21/21
+1	0.42	0.03	1.8	0.64	21/21	2.10	0.13	9.7	1.45	21/21
+2	0.49	0.05	2.0	0.77	21/21	1.95	0.12	9.0	1.72	21/21
+3	0.18	0.02	2.4	0.73	20/20	3.82	0.32	9.3	1.67	20/20
+4	0.20	0.03	2.2	0.79	20/20	2.19	0.13	8.6	1.94	20/20
+5	0.17	0.03	1.9	0.72	20/20	2.13	0.11	9.5	1.77	20/20
+6	0.19	0.03	2.2	0.89	19/19	0.95	0.07	9.8	2.01	19/19
+7	0.11	0.02	1.9	0.57	18/18	1.84	0.13	8.6	1.87	18/18
+8	0.24	0.04	2.2	0.73	18/18	2.91	0.25	8.6	1.81	18/18
+9	0.18	0.04	2.8	0.85	18/18	0.41	0.04	7.6	2.11	18/18
+10	0.22	0.04	2.6	0.68	17/17	0.54	0.05	7.7	1.70	17/17
+11	0.26	0.05	2.6	0.90	16/16	6.46	0.46	9.8	2.82	16/16
+12	0.30	0.04	3.0	1.14	15/15	1.95	0.25	10.3	2.57	15/15